Unit - II: Partial Differentiation

Tutorial Sheet - 1

1. If $z(x + y) = x^2 + y^2$ then show that $\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left(1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)$.
2. If $u = (x^2 + y^2 + z^2)^{-1/2} = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$ then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$
3. If $u = log(x^3 + y^3 + z^3 - 3xyz)$ then show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x + y + z)^2}$
4. If $v = \log(x^2 + y^2 + z^2)$ then $(x^2 + y^2 + z^2)\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2}\right) = 2$
5. If $x = r\cos\theta$, $y = r\sin\theta$ then prove the following $(i)\frac{\partial r}{\partial x} = \frac{\partial x}{\partial r}$ $(ii)\ \frac{1}{r}\frac{\partial x}{\partial \theta} = r\frac{\partial \theta}{\partial x}$

$(iii)\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2 = 1\ (iv)\ \frac{\partial^2 r}{\partial x^2} + \frac{\partial^2 r}{\partial y^2} = \frac{1}{r}\left[\left(\frac{\partial r}{\partial x}\right)^2 + \left(\frac{\partial r}{\partial y}\right)^2\right] (v)\ \frac{\partial^2 \theta}{\partial x^2} + \frac{\partial^2 \theta}{\partial y^2} = 0$

6. If $r^2 = x^2 + y^2 + z^2$ and $v = r^m$ then prove that
   $v_{xx} + v_{yy} + v_{zz} = m(m + 1)r^{m-2}$
7. If $z = f(x + ct) + \phi(x - ct)$ then show that $\frac{\partial^2 z}{\partial t^2} = c^2\frac{\partial^2 z}{\partial x^2}$
8. If $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$ then $Show\ that$: $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 0$